AgriTrack: Revolutionizing Agricultural Supply Chains

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BSc Information Technology Capstone

The Core Problem

- Smallholder farmers face predatory middlemen taking majority profits.
- Lack of real-time market data leads to post-harvest losses.
- Inefficient supply chains cause delays between farm and commercial buyers.
- Poor data tracking prevents farmers from securing agricultural credit.

The Solution: AgriTrack

- A direct-to-buyer agricultural marketplace and supply chain tracker.
- Connects rural smallholder farmers directly to commercial buyers.
- Provides real-time inventory management for crops.
- Ensures transparent pricing and logistics tracking.

Technical Architecture

- Frontend: Kotlin Multiplatform & React Native for cross-platform access.
- Backend: NestJS providing robust API architecture.
- Database: WatermelonDB for offline-first local data synchronization.
- Deployment: Local integration capabilities for rural areas with low connectivity.

Key Innovative Features

- Offline-first synchronization allows farmers to log yields without immediate internet.
- Direct messaging module between buyers and farmers.
- Geospatial tracking of produce from harvest to delivery.
- Automated smart contracts and digital invoicing.

Impact on the IT & Agriculture Sectors

- Demonstrates the power of decentralized mobile applications.
- Bridges the gap between physical agriculture and digital economies.
- Empowers farmers economically by reclaiming profit margins.
- Scalable architecture can be adapted to other supply chains in Africa.